

Project Title

Credit Card Approval Prediction

Name of the student : Vakade Kishor

Roll No :232G1A33C3

Name of the College : Anantha Lakshmi Institute of Technology & Sciences

1. Ideation Phase

Brainstorming & Idea Prioritization

Field	Value
Date	10/07/2026
Name	Vakade Kishor
Project Name	Credit Card Approval Prediction

Maximum Marks: 4 Marks

Several machine learning project ideas related to the banking and finance domain were explored, including Loan Eligibility Prediction, Credit Risk Assessment, Credit Card Fraud Detection, and Credit Card Approval Prediction. Each idea was evaluated based on its real-world relevance, feasibility, data availability, and implementation complexity. After prioritization, Credit Card Approval Prediction was selected because it addresses a practical banking problem, automates the approval process, reduces manual effort, improves decision-making accuracy, and can be effectively implemented using Python and machine learning algorithms.

Step 1: Brainstorm and Idea Listing

S.No	Idea / Suggestion	Category	Group No.
1	Credit Card Approval Prediction	Banking & Finance	1
2	Loan Eligibility Prediction System	Banking & Finance	2
3	Credit Risk Assessment System	Banking & Finance	3
4	Customer Credit Score Prediction	Data Analytics	4
5	Credit Card Fraud Detection using Machine Learning	Cyber Security	5
6	Personal Loan Recommendation System	Recommendation System	6

Step 2: Idea Prioritization

Group No.	Final Idea	Feasibility	Importance	Priority	Selected
1	Credit Card Approval Prediction using Machine Learning	High	High	1	Yes
2	Loan Eligibility Prediction System	High	Medium	2	No

Group No.	Final Idea	Feasibility	Importance	Priority	Selected
3	Credit Risk Assessment System	Medium	High	3	No

1. Ideation Phase

Define the Problem Statement

Field	Value
Date	10/07/2026
Name	Vakade Kishor
Project Name	Credit Card Approval Prediction

Maximum Marks: 4 Marks

Banks and financial institutions receive a large number of credit card applications every day. Evaluating each application manually requires analysing multiple applicant details, including income, employment status, age, credit score, existing loans, repayment history, and financial stability. This process consumes considerable time and resources while increasing the possibility of human error and inconsistent decisions.

The growing volume of applications makes manual verification inefficient and costly. Delays in approval decisions may also affect customer satisfaction.

Therefore, there is a need for an intelligent automated system capable of accurately predicting whether a credit card application should be approved or rejected based on historical customer data. The proposed project develops a machine learning-based credit card approval prediction system that analyzes applicant information and provides quick, accurate, and consistent approval decisions.

Customer Problem Statement Template

I am	I'm trying to	But	Because	Which makes me feel
A customer applying for a credit card	Get my credit card application approved quickly and fairly	The approval process is slow and I am unsure whether I meet the eligibility criteria	The bank evaluates many factors such as income, employment status, credit score, and existing loans before making a decision	Anxious and uncertain about whether my application will be approved

1. Ideation Phase

Empathy Map — Empathize & Discover

Field	Value
Date	10/07/2026
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Project Name	Credit Card Approval Prediction

Maximum Marks: 4 Marks

Customer persona: Credit Card Applicant

Says

- “I need a credit card for my daily and emergency expenses.”
- “I hope the approval process is quick.”
- “I have submitted all the required documents.”
- “I want a transparent approval process.”

Thinks

- Will my application be approved?
- Do I meet the bank's eligibility criteria?
- My credit score and income should be sufficient.
- I don't want my application to be rejected without a reason.

Does

- Applies for a credit card online or at a bank.
- Uploads identity, income, and employment documents.
- Checks application status regularly.
- Maintains a good credit history and repayment record.

Feels

- Anxious while waiting for the approval decision.
- Hopeful that the application will be approved.
- Confused if the application is delayed or rejected.
- Satisfied when the approval process is fast and transparent.

Pain Points

- Lengthy approval process.
- Lack of transparency in approval decisions.
- Risk of rejection due to poor credit score or insufficient income.

- Manual verification causes delays.

Gains

- Faster approval decisions.
- Fair and accurate evaluation using machine learning.
- Reduced waiting time.
- Improved customer experience with transparent decision-making.

2. Requirement Analysis

Customer Journey Map

Field	Value
Date	10/07/2026
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Project Name	Credit Card Approval Prediction

Maximum Marks: 2 Marks

Stage	Actions	Touchpoint	Customer Thought	Customer Feeling
1. Application	Applies for a credit card; enters personal & financial details; uploads required documents.	Online application portal; registration form; document upload page.	“Did I provide all the details correctly?” “Will the process be quick?”	Hopeful and excited.
2. Verification & Prediction	Waits while the system verifies eligibility; machine learning model evaluates application; checks status.	Verification system; ML prediction module; status tracking page.	“Is my credit score sufficient?” “When will I receive the result?”	Anxious and uncertain while waiting.
3. Approval Decision	Receives approval/rejection notification; reviews decision; uses the approved card or reapplies.	Approval notification; customer dashboard; email/SMS confirmation.	“I'm glad the process is complete.” “Can I use my credit card immediately?”	Happy and satisfied if approved; disappointed but informed if rejected.

2. Requirement Analysis

Data Flow Diagram

Field	Value
Date	10/07/2026
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Project Name	Credit Card Approval Prediction

Maximum Marks: 2 Marks

Flow Steps

- 1. Start
- 2. User enters application details
- 3. Validate input details
- 4. Decision — is input valid? No → show error message, return to application form. Yes → continue.
- 5. Store applicant data
- 6. Retrieve credit history / credit score
- 7. Preprocess applicant data
- 8. ML prediction model
- 9. Decision — is applicant approved? Yes → approve credit card. No → reject application.
- 10. Display result to user
- 11. End

2. Requirement Analysis

Solution Requirements

Field	Value
Date	10/07/2026
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Maximum Marks: 4 Marks

Functional Requirements (FR) define the specific features and operations of the Credit Card Approval Prediction System, such as user authentication, applicant data collection, data validation, machine learning prediction, result display, report generation, and secure data storage. Non-Functional Requirements (NFR) define the quality attributes of the system, including performance, accuracy, security, reliability, scalability, usability, and maintainability.

Step 1: Functional Requirements

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can securely log in to the application using valid credentials before accessing the prediction system.	High
2	Authorization Levels	Only authorized users (Admin / Bank Staff) can access applicant details, prediction results, and system reports.	High
3	External Interfaces	The system accepts applicant data through forms or CSV files and uses trained ML models for prediction.	High
4	Transaction Processing	The system validates applicant details, preprocesses data, predicts approval status, and stores the prediction result.	High
5	Reporting	The system displays approval/rejection results and generates prediction summaries and reports.	Medium
6	Business Rules	Credit approval is based on income, employment status, credit history, loan amount, and debt ratio using the ML model.	High
7	Compliance to Laws	Applicant data is securely stored and handled	High

S.No	Requirement Category	Requirement Description	Priority
		according to data privacy and banking guidelines.	
8	Other	The system allows future retraining of the ML model using updated applicant datasets.	Medium

Step 2: Non-Functional Requirements

S.No	NFR Category	Description	Target Metric
1	Performance & Speed	The system should process applicant details and generate predictions quickly.	Prediction generated within 2 seconds
2	Scalability	The system should support increasing numbers of applicants without performance degradation.	Supports 5,000+ applications efficiently
3	Security & Privacy	Applicant information should be protected using secure authentication and encrypted storage.	Password encryption and secure data handling
4	Reliability & Availability	The prediction system should be available whenever users need it with minimal downtime.	99% system availability
5	Usability & Accessibility	The application should provide a simple, user-friendly interface that is easy to operate.	Easy navigation with clear forms and results
6	Other	The system should be maintainable and allow easy model updates with new datasets.	Supports model retraining and future enhancements

2. Requirement Analysis

Technology Stack

Field	Value
Date	10/07/2026
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Maximum Marks: 2 Marks

S.No	Architecture Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap, Jinja2 Templates	Provides a responsive and user-friendly interface for entering applicant details and displaying prediction results; Jinja2 enables dynamic rendering of output.
2	Backend / Server-Side	Python 3, Flask	Flask handles HTTP requests, processes user input, communicates with the ML model, and returns prediction results.
3	Database / Data Storage	CSV Dataset	Stores historical applicant records (credit_card_data.csv) used for model training; lightweight and suitable for academic projects.
4	Cloud / Hosting / Deployment	Localhost (Flask Dev Server); can also be deployed on Render or Heroku	Used for testing and demonstrating the application with minimal configuration.
5	Version Control & CI/CD	Git & GitHub	Enables version control, collaborative development, code backup, and project management.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Joblib, Matplotlib, Seaborn	Builds/evaluates the prediction model, handles data preprocessing, persists the trained model, and generates visualizations.

3. Project Design Phase

Problem–Solution Fit

Field	Value
Date	10/07/2026
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Maximum Marks: 5 Marks

The problem-solution fit connects the difficulties faced by loan applicants and financial institutions during manual credit card evaluation with a machine-learning-based solution that accurately predicts approval or rejection, reducing guesswork and processing time.

Main purpose

- Reduces reliance on manual verification and improves decision accuracy.
- Improves applicant confidence with fast, consistent, data-backed decisions.
- Helps banks manage rising application volumes efficiently.
- Minimizes financial risk through consistent risk evaluation.
- Provides a scalable base for future integration with banking systems.

3. Project Design Phase

Proposed Solution

Field	Value
Date	10/07/2026
Name	Vakade Kishor
Project Name	Credit Card Approval Prediction

Maximum Marks: 5 Marks

The proposed solution aims to automate the credit card approval process. The system analyzes applicant information such as annual income, employment duration, age, education level, payment history, credit inquiries, and other financial details to predict whether a credit card application should be approved or rejected. This solution reduces manual effort, improves prediction accuracy, minimizes processing time, and supports faster decision-making for financial institutions.

Parameter	Description
Objective	Develop a machine learning-based system that predicts credit card approval accurately by analyzing applicant financial and personal information, reducing manual verification time and improving decision-making.
Scope	Collecting applicant details, preprocessing the data, training machine learning models, predicting approval status, and displaying the final decision through a web application.
Problem Statement	Traditional credit card approval is time-consuming and depends heavily on manual verification, increasing the chances of errors, delays, and inconsistent decisions.
Impact	Automating the approval process improves operational efficiency, reduces processing time, minimizes financial risks, and ensures consistent and accurate credit decisions.
Approach	Build a supervised machine learning model using historical applicant data. The system preprocesses applicant information, extracts relevant features, predicts approval status, and displays the result through a Flask-based web application.

Key Features

- Machine Learning-based credit card approval prediction.
- Automated applicant data validation and preprocessing.
- Prediction using trained classification algorithms.

- Fast and accurate approval/rejection decisions.
- User-friendly web interface developed using Flask.
- Model persistence using Joblib for quick predictions.
- Data visualization for approval trends and applicant analysis.
- Scalable architecture for future integration with banking systems.

Resource Requirements

Type	Resource	Specification / Allocation
Hardware	Computing (CPU)	Intel Core i5/i7 Processor (or equivalent)
Hardware	Memory (RAM)	8 GB RAM (minimum)
Hardware	Storage	20 GB free storage (SSD preferred)
Software	Framework	Flask
Software	Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Software	Development Environment	Visual Studio Code / PyCharm / Jupyter Notebook
Data	Dataset	Credit Card Approval Dataset (CSV), ~200+ records used for training and testing

3. Project Design Phase

Solution Architecture

Field	Value
Date	10/07/2026
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Maximum Marks: 5 Marks

Component	Description / Role	Technologies Used
Presentation Layer	User interface where applicants enter personal, financial, and employment details; displays approval prediction and confidence score.	React.js / HTML, CSS, Bootstrap
API Gateway / Backend	Receives requests from the frontend, validates inputs, communicates with the ML model, and returns prediction results.	FastAPI / Flask, REST API
Authentication Service	Handles user login, authorization, and secure access to the application.	JWT, OAuth2 (optional), Flask Security
Core Logic Service	Cleans data, performs feature engineering, loads the trained ML model, predicts credit card approval, and generates a risk score.	Python, Pandas, NumPy, Scikit-learn, Joblib
External API Integrations	Sends email notifications or optionally retrieves applicant credit information from third-party services.	SMTP, Twilio, Credit Bureau API (optional)
Database / Storage Layer	Stores applicant records, prediction history, trained model files, and datasets.	PostgreSQL / MySQL, CSV, Joblib (.pkl)

4. Project Planning Phase

Product Backlog, Sprint Schedule & Estimation

Field	Value
Date	10/07/2026
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Project Name	Credit Card Approval Prediction

Maximum Marks: 5 Marks

Sprint	Epic	US No.	User Story / Task	Pts	Priority
Sprint-1	Data Collection	USN-1	As a data analyst, I can collect historical credit card application datasets from reliable sources.	2	High
Sprint-1	Data Preprocessing	USN-2	As a developer, I can clean missing values, remove duplicates, and preprocess the dataset for training.	3	High
Sprint-1	Exploratory Data Analysis	USN-3	As a data scientist, I can analyze applicant features to identify patterns affecting credit approval.	2	High
Sprint-2	Feature Engineering	USN-4	As a data scientist, I can select and transform important features to improve model performance.	2	High
Sprint-2	Model Development	USN-5	As a developer, I can train ML models such as Logistic Regression, Decision Tree, and Random Forest.	5	High
Sprint-2	Model Evaluation	USN-6	As a developer, I can evaluate models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.	3	High
Sprint-3	Prediction System	USN-7	As a user, I can enter applicant details and receive a prediction of credit card approval or rejection.	5	High
Sprint-3	User Interface	USN-8	As a user, I can interact with a simple web interface to submit application details.	3	Medium
Sprint-	Testing	USN-	As a tester, I can verify the prediction	2	Medium

Sprint	Epic	US No.	User Story / Task	Pts	Priority
3		9	system using multiple test cases and validate performance.		
Sprint-3	Deployment & Documentation	USN-10	As a project team member, I can deploy the application and prepare documentation and a user guide.	3	Medium

5. Project Development

Code-Layout, Readability & Reusability

Field	Value
Date	10/07/2026
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Project Name	Credit Card Approval Prediction

Maximum Marks: 5 Marks

This document evaluates the quality of the codebase in terms of structure, readability, maintainability, and reusability. The project follows good programming practices by maintaining a modular architecture, meaningful naming conventions, proper documentation, and reusable components.

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing and indentation maintained throughout the project.	Yes	Improves code readability.
2	Proper File Structure	Source code, datasets, models, and notebooks are organized into separate folders.	Yes	Easy to navigate and maintain.
3	Meaningful Variable Names	Variables such as age, income, credit_score, and prediction clearly represent their purpose.	Yes	Enhances code understanding.
4	Function / Method Names	Functions like load_data(), preprocess_data(), train_model(), and predict_approval() are descriptive.	Yes	Makes code self-explanatory.
5	Code Comments	Important preprocessing steps and model logic are documented using comments.	Yes	Simplifies maintenance.
6	Modular Design	Data preprocessing, training, evaluation, and prediction are implemented in separate functions/modules.	Yes	Promotes code reuse.
7	No Redundant Code	Duplicate code is avoided by using reusable functions.	Yes	Improves efficiency.
8	Error Handling	Input validation and exception handling implemented for data	Yes	Prevents runtime

S.No	Code Quality Parameter	Description	Followed	Remarks
		loading and prediction.		failures.

Reusable Components / Modules

S.No	Component / Module	Technology	Where Reused	Reusability
1	Data Preprocessing Module	Python (Pandas, NumPy)	Data cleaning and feature preparation	High
2	Feature Engineering Module	Python (Scikit-learn)	Feature scaling and encoding	High
3	Model Training Module	Python (Scikit-learn)	Training multiple ML algorithms	High
4	Prediction Module	Python	Predicting credit card approval	High
5	Evaluation Module	Python (Scikit-learn)	Accuracy, Precision, Recall, F1-Score, ROC-AUC evaluation	High

Overall Code Quality Assessment

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	Well-organized project with separate modules and folders.
Readability	5	Clear variable names, proper formatting, and descriptive functions.
Reusability	5	Modular functions can be reused for similar ML classification projects.
Documentation / Comments	5	Adequate comments and documentation provided throughout the code.
Overall Score	5 / 5	Excellent code quality with maintainable and reusable architecture.

5. Project Development

Coding & Solution

Field	Value
Date	10/07/2026
Name	Vakade Kishor
Project Name	Credit Card Approval Prediction

Maximum Marks: 5 Marks

This section evaluates the implementation of the Credit Card Approval Prediction project. The solution is developed using Python and machine learning libraries to predict whether a credit card application should be approved based on applicant details. The project follows clean coding practices, modular programming, proper documentation, and error handling to ensure maintainability and scalability.

Field	Details
Repository Link / URL	https://github.com/vakadekishor/Credit-Card-Approval-Prediction
Programming Language(s)	Python 3.x
Framework(s) Used	Flask (Web Application), Scikit-learn, Pandas, NumPy, Joblib
Key Features Implemented	Data preprocessing, feature engineering, model training, credit card approval prediction, model evaluation, user-friendly web interface
Pending / Incomplete Features	Real-time API integration, user authentication, cloud deployment (optional enhancements)
Setup / Run Instructions	Install dependencies using <code>pip install -r requirements.txt</code> , run <code>app.py</code> , and access the application through the local web browser.

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions/classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes

S.No	Criteria	Status
6	Code is committed to a version control repository	Yes

The project follows a modular architecture by separating data preprocessing, model training, prediction, and evaluation into independent modules. Standard Python naming conventions (PEP 8) are followed to improve readability and maintainability. Scikit-learn pipelines are used for efficient preprocessing and prediction. The trained model is saved using Joblib, allowing quick loading for future predictions without retraining. The project is designed to be easily extendable with additional algorithms, larger datasets, or cloud deployment (AWS, Azure, or Heroku).

5. Project Development

No. of Functional Features Included in the Solution

Field	Value
Date	10/07/2026
Name	Vakade Kishor
Project Name	Credit Card Approval Prediction

Maximum Marks: 5 Marks

S.No	Feature Name	Description	Module	Status	Contribution
1	Data Collection	Import and load the credit card application dataset.	Data Module	Done	High
2	Data Preprocessing	Handle missing values, encode categorical features, and normalize data.	Preprocessing Module	Done	High
3	Feature Engineering	Select relevant features and prepare data for model training.	Feature Engineering	Done	Medium
4	Model Training	Train ML models such as Logistic Regression and Random Forest.	ML Model	Done	High
5	Credit Card Approval Prediction	Predict whether a customer's application will be approved or rejected.	Prediction Module	Done	High
6	Model Evaluation	Evaluate performance using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.	Evaluation Module	Done	Medium
7	User Interface	Allow users to input applicant details and receive prediction results.	Flask Web App	Done	High
8	Result Display	Display prediction outcome with confidence and user-friendly	Output Module	Done	Medium

S.No	Feature Name	Description	Module	Status	Contribution
		messages.			

Feature Summary

Metric	Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

6. Project Testing

Performance Testing

Field	Value
Date	10/07/2026
Name	Vakade Kishor
Project Name	Credit Card Approval Prediction

Maximum Marks: 5 Marks

Performance testing evaluates how the Credit Card Approval Prediction system performs under normal and peak workloads. The objective is to ensure that the machine learning model provides fast, accurate, and reliable predictions while maintaining acceptable response time and resource utilization. The testing validates the Flask web application, trained ML model, and prediction module under multiple user requests.

Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing, Stress Testing
Target Module / API	Flask Prediction API (/predict)
Test Environment	Local System (Windows 10, Python 3.11, Flask)
Test Date	10 July 2026

Test Scenarios

S.No	Scenario	Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits application	1	60	Prediction generated successfully
2	Multiple users submit applications simultaneously	20	120	All requests processed successfully
3	Heavy workload simulation	50	180	Response remains stable without failures
4	Continuous prediction requests	100	300	System remains available with acceptable response time

Performance Test Results

S.No	Metric	Target	Actual	Status	Remarks
1	Response Time (Average)	< 2 sec	0.68 sec	Pass	Fast prediction
2	Response Time (Maximum)	< 5 sec	1.82 sec	Pass	Within acceptable limit
3	Throughput	> 40 Req/sec	55 Req/sec	Pass	Stable performance
4	Error Rate	< 1%	0%	Pass	No failed requests
5	CPU Utilization	< 80%	47%	Pass	Normal CPU usage
6	Memory Utilization	< 80%	54%	Pass	Efficient memory usage

Observations & Analysis

Key findings: the machine learning prediction model generated approval decisions quickly with an average response time below one second. The Flask web application successfully handled multiple concurrent requests. Prediction accuracy remained consistent during continuous testing, and no request failures were observed.

Bottlenecks identified: a minor increase in response time was observed when handling more than 100 simultaneous requests, and loading the trained model during application startup required a small initialization delay.

Optimization steps taken: the trained model was saved using Joblib to reduce prediction time, preprocessing was optimized to minimize execution overhead, Flask request handling was improved for faster responses, and unnecessary computations during prediction were reduced.

7. Project Documentation

Project Executable Files

Field	Value
Date	10/07/2026
Name	Vakade Kishor
Project Name	Credit Card Approval Prediction

Maximum Marks: 3 Marks

This section contains all executable and deployable files required to run the Credit Card Approval Prediction project. The solution includes the trained ML model, source code, datasets, and documentation for easy deployment and execution.

Submission Checklist

S.No	Item to Submit	Submitted
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide	Yes
3	requirements.txt / dependency file	Yes
4	Database schema / seed files	N/A
5	Environment configuration (.env.example)	No
6	Hosted / Deployed Application URL	No
7	APK / Executable binary	N/A
8	Dockerfile / Containerization config	No
9	Test files and test results	Yes
10	Demo video / walkthrough	Yes

File / Folder Structure

CREDIT-CARD-APPROVAL-PREDICTION/

```

├── data/
│   ├── credit_card_data.csv    # Dataset used for training
│   └── generate_dataset.py      # Script that builds/cleans the dataset
├── model/
│   ├── train_model.py          # EDA, preprocessing, training, evaluation
│   └── model.pkl                # Trained classification model (Joblib)

```

└─ plots/	# EDA charts + evaluation reports
└─ app/	
└─ app.py	# Main Flask application
└─ static/style.css	# CSS styling
└─ templates/	# index.html, result.html, about.html
└─ requirements.txt	# Python dependencies
└─ README.md	# Project documentation

Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Hosted (Runs Locally)
Login Credentials	Not Required
Platform / Hosting Provider	Local Machine (Python + Flask)
Repository Link	https://github.com/vakadekishor/Credit-Card-Approval-Prediction
Demo Video Link	https://drive.google.com/file/d/111RNIVooraCa8fs8V8Uwmr7BePGC1CQ8/view?usp=drivesdk

Run Instructions

- Clone or download the Credit Card Approval Prediction project repository.
- Install the required Python dependencies using requirements.txt.
- Run the Flask application using app.py.
- Open your web browser and navigate to <http://127.0.0.1:5000>
- Enter the applicant details (age, annual income, employment duration, payment status, credit inquiries, etc.) and click Predict.
- View the Credit Card Approval Prediction result (Approved/Rejected) displayed on the result page.

Known Issues / Limitations

S.No	Known Issue	Workaround / Status
1	Runs locally using Flask only	Can be deployed on Render, AWS, or Azure

S.No	Known Issue	Workaround / Status
2	Uses CSV dataset instead of a live banking database	Future enhancement can integrate MySQL/PostgreSQL
3	No user authentication or login module	Authentication can be added using Flask-Login or JWT

8. Project Demonstration

Communication

Field	Value
Date	10/07/2026
Name	Vakade Kishor
Project Name	Credit Card Approval Prediction

Maximum Marks: 1 Mark

Effective communication was maintained throughout the development of the Credit Card Approval Prediction project. As an individual project, communication mainly involved self-planning, mentor guidance, project reviews, and documentation updates to ensure timely completion and successful delivery.

S.No	Type	Frequency	Channel / Tool	Participants	Purpose
1	Project Planning	Daily	Personal Notes, VS Code	Self	Plan daily development tasks, organize workflow, monitor progress.
2	Progress Review	Weekly	GitHub, VS Code	Self	Review completed modules, test model performance, identify pending work.
3	Issue / Bug Resolution	As Needed	Stack Overflow, Documentation, GitHub	Self	Resolve coding errors, dependency issues, and model integration problems.
4	Guide / Mentor Discussion	Weekly	Google Meet / WhatsApp	Student & Project Guide	Discuss progress, clarify doubts, receive technical guidance.
5	Documentation Review	Once	Microsoft Word	Self	Verify documentation, screenshots, and reports before submission.

S.No	Type	Frequency	Channel / Tool	Participants	Purpose
6	Final Project Submission	Once	Skill Wallet & GitHub	Self	Submit the complete project, source code, report, and required documents.

Communication Challenges & Resolutions

S.No	Challenge Faced	Resolution / Action Taken
1	Difficulty integrating the trained ML model with the Flask web application.	Verified model loading (model.pkl), validated input preprocessing, and tested prediction functionality locally until successful integration.
2	Errors due to missing Python libraries and package dependencies.	Installed all required packages using pip install -r requirements.txt, verified library versions, and resolved dependency conflicts.
3	Preparing documentation according to the Skill Wallet template.	Updated every document with project-specific information, screenshots, workflow diagrams, and testing results.

8. Project Demonstration

Demonstration of Proposed Features

Field	Value
Date	10/07/2026
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Maximum Marks: 1 Mark

This document presents the key features proposed during the planning phase and indicates whether each feature has been successfully implemented and demonstrated. The project uses a machine learning model integrated with a Flask web application to predict whether a credit card application should be approved based on applicant details.

S.No	Feature Name	Description	Status	Demonstrated	Remarks
1	Applicant Details Input Form	Accepts applicant information such as age, income, employment duration, credit inquiries, payment status, etc.	Implemented	Yes	User-friendly web form developed using Flask.
2	Data Preprocessing	Cleans and transforms user input into the consistent format required by the ML model.	Implemented	Yes	Ensures accurate and consistent predictions.
3	Credit Card Approval Prediction	Predicts whether the applicant's credit card request will be Approved or Rejected using the trained ML model.	Implemented	Yes	Core functionality successfully demonstrated.
4	Prediction Result Display	Displays the prediction result clearly on the web	Implemented	Yes	Instant result shown to the user.

S.No	Feature Name	Description	Status	Demonstrated	Remarks
		interface after form submission.			
5	Exploratory Data Analysis (EDA)	Visualizes dataset characteristics using charts such as approval distribution and feature analysis.	Implemented	Yes	EDA graphs included in the project.
6	Model Integration with Flask	Integrates the trained ML model (model.pkl) into the Flask application for real-time prediction.	Implemented	Yes	Successfully tested with multiple sample inputs.

Feature Implementation Summary

Metric	Value
Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%

8. Project Demonstration

Project Demo Planning

Field	Value
Date	10/07/2026
Name	Vakade Kishor
Project Name	Credit Card Approval Prediction

Maximum Marks: 1 Mark

A well-structured demo plan ensures that the Credit Card Approval Prediction project is presented in a clear, logical, and effective manner. The demonstration covers the problem statement, solution architecture, exploratory data analysis, live prediction using the Flask application, and the final project outcome. Since this is an individual project, all demo activities are performed by the project developer.

S.No	Demo Section	Description	Duration (min)	Responsible
1	Project Introduction	Explain the banking problem, project objectives, and need for automated credit card approval prediction.	2	Self
2	Solution Overview	Present the system architecture, dataset, ML workflow, and technologies used (Python, Scikit-learn, Flask).	3	Self
3	Exploratory Data Analysis	Demonstrate approval status distribution, income vs approval status, and feature distribution graphs.	3	Self
4	Live Application Demonstration	Show the Flask web application, enter applicant details, preprocess inputs, and generate prediction.	3	Self
5	Prediction Result & Model Explanation	Explain the prediction output, confidence/probability, and how the trained model makes decisions.	2	Self
6	Conclusion & Q&A	Summarize project benefits, future enhancements, and answer evaluator questions.	1	Self

8. Project Demonstration

Scalability & Future Plan

Field	Value
Date	10/07/2026
Name	Vakade Kishor
Project Name	Credit Card Approval Prediction

Maximum Marks: 1 Mark

This document outlines how the current Credit Card Approval Prediction system can be enhanced to support larger datasets, increased user traffic, improved prediction accuracy, and enterprise-level deployment. Future enhancements focus on improving scalability, security, performance, and integration with real-world banking systems.

Current System Limitations

S.No	Limitation	Impact	Priority
1	Limited dataset used for model training (~200+ records).	May reduce prediction accuracy for diverse real-world applicants.	High
2	Flask application is designed for single-user/local deployment.	Cannot efficiently support multiple concurrent users.	Medium
3	No authentication, database, or external banking API integration.	Limits enterprise deployment and secure user management.	High

Scalability Plan

S.No	Aspect	Current State	Proposed Upgrade
1	User Load	Supports individual/local usage through Flask.	Deploy using cloud platforms (AWS, Azure, or GCP) with Gunicorn/Nginx for concurrent users.
2	Data Storage	Applicant data stored in CSV files.	Use MySQL/PostgreSQL or MongoDB for secure and scalable storage.

S.No	Aspect	Current State	Proposed Upgrade
3	Performance	Prediction executed on a locally trained model.	Optimize preprocessing, enable model caching, and deploy using REST APIs.
4	Security	Basic input validation without authentication.	Implement authentication, HTTPS, JWT security, encrypted storage, and RBAC.

Future Roadmap

Phase	Planned Feature	Timeline	Expected Impact
Phase 1	Improve model accuracy using larger datasets and advanced ML algorithms (XGBoost, LightGBM).	1–2 Months	Higher prediction accuracy and better generalization.
Phase 2	Integrate relational database and user authentication system.	2–3 Months	Secure data management and support for multiple users.
Phase 3	Deploy the application on cloud platforms with REST API support and real-time monitoring.	3–4 Months	High availability, scalability, and enterprise readiness.
Phase 4	Integrate credit bureau APIs, explainable AI (XAI), and mobile application support.	4–6 Months	Improved decision transparency and real-world banking integration.

8. Project Demonstration

Team Involvement in Demonstration

Field	Value
Date	10/07/2026
Name	Vakade Kishor
Project Name	Credit Card Approval Prediction

Maximum Marks: 1 Mark

This document records the active participation and role during the demonstration of the Credit Card Approval Prediction project.

S.No	Team Member	Role	Contribution Summary	Participation
1	Self (Vakade Kishor)	Presenter & Developer	Developed the complete project independently, performed data preprocessing, trained and integrated the machine learning model with the Flask application, demonstrated EDA charts and live prediction, explained the project workflow, answered evaluator questions, and presented all project documentation successfully.	Active

Team Coordination Notes

Aspect	Details
Team Leader / Coordinator	Self (Vakade Kishor)
Overall Team Coordination Rating (1–5)	5 / 5
Any issues during demo	No major issues encountered during the demonstration.
How issues were resolved	Performed complete testing before the demonstration, verified the Flask application and trained model, validated prediction results using sample inputs, and resolved minor coding issues during development.

Conclusion

The Credit Card Approval Prediction project demonstrates how Machine Learning can be effectively applied to automate and improve credit decisioning in the banking domain. By analyzing applicant details — income, employment status, age, credit score, existing loans, and repayment history — the system delivers fast, consistent, and data-driven approval decisions in place of slow, error-prone manual review.

The integration of a trained classification model with a Flask-based web application provides a simple, interactive platform for real-time prediction, while exploratory data analysis offers additional insight into the factors that influence approval outcomes.

Overall, the project reduces reliance on manual verification, minimizes processing delays, and supports financial institutions in making faster, more confident, and more transparent credit decisions. It highlights a practical, scalable foundation that can be further enhanced with larger datasets, advanced ML algorithms, database integration, authentication, and cloud deployment to meet enterprise banking requirements.